Namespace Dear_ImGui_Sample.Backends

Classes

AssetLoader

BedrockImGUICompatability

ImguiImplOpenGL3

Class AssetLoader

Namespace: Dear_ImGui_Sample.Backends

Assembly: Osmium-Editor.dll

```
public static class AssetLoader
```

Inheritance

object ← AssetLoader

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Font(string, float)

```
public static ImFontPtr Font(string path, float size)
```

Parameters

`path` string

`size` float

Returns

ImFontPtr

Image(string)

```
public static Texture Image(string path)
```

Parameters

`path` string

Returns

Texture

Class BedrockImGUICompatability

Namespace: Dear_ImGui_Sample.Backends

Assembly: Osmium-Editor.dll

```
public static class BedrockImGUICompatability
```

Inheritance

object ← BedrockImGUICompatability

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ScreenSize

```
public static Vector2 ScreenSize
```

Field Value

Vector2

Methods

Incorporate()

```
public static void Incorporate()
```

Class ImguiImplOpenGL3

Namespace: Dear_ImGui_Sample.Backends

Assembly: Osmium-Editor.dll

```
public static class ImguiImplOpenGL3
```

Inheritance

object ← ImguiImplOpenGL3

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Init(bool)

```
public static bool Init(bool useKHRDebugIfAvailable = true)
```

Parameters

`useKHRDebugIfAvailable` bool

Returns

bool

NewFrame()

```
public static void NewFrame()
```

RenderDrawData(ImDrawDataPtr)

```
public static void RenderDrawData(ImDrawDataPtr drawData)
```

Parameters

`drawData` ImDrawDataPtr

SetupRenderState(ImDrawDataPtr, int, int, int)

```
public static void SetupRenderState(ImDrawDataPtr drawData, int fbWidth, int
fbHeight, int vao)
```

Parameters

`drawData` ImDrawDataPtr

`fbWidth` int

`fbHeight` int

`vao` int

Shutdown()

```
public static void Shutdown()
```

Namespace Dear_ImGui_Sample.Backends.Types

Classes

Texture

Class Texture

Namespace: Dear_ImGui_Sample.Backends.Types

Assembly: Osmium-Editor.dll

```
public record Texture : IEquatable<Texture>
```

Inheritance

object ← Texture

Implements

IEquatable<Texture>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Texture(nint, int, int)

```
public Texture(nint Handle, int Width, int Height)
```

Parameters

`Handle` nint

`Width` int

`Height` int

Properties

Handle

```
public nint Handle { get; init; }
```

Property Value

nint

Height

```
public int Height { get; init; }
```

Property Value

int

Width

```
public int Width { get; init; }
```

Property Value

int

Operators

implicit operator nint(Texture)

```
public static implicit operator nint(Texture texture)
```

Parameters

`texture` Texture

Returns

nint

Namespace OsmiumBedrock

Classes

Bedrock

Window

Class Bedrock

Namespace: OsmiumBedrock

Assembly: Osmium-Bedrock.dll

```
public static class Bedrock
```

Inheritance

object ← Bedrock

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

DeltaTime

```
public static float DeltaTime
```

Field Value

float

window

```
public static readonly Window window
```

Field Value

Window

Methods

Run()

```
public static void Run()
```

Events

Draw

```
public static event Action Draw
```

Event Type

Action

DrawFinalizer

```
public static event Action DrawFinalizer
```

Event Type

Action

DrawInitializer

```
public static event Action DrawInitializer
```

Event Type

Action

Load

```
public static event Action Load
```

Event Type

Action

Unload

```
public static event Action Unload
```

Event Type

Action

Update

```
public static event Action Update
```

Event Type

Action

UpdateFinalizer

```
public static event Action UpdateFinalizer
```

Event Type

Action

UpdateInitializer

```
public static event Action UpdateInitializer
```

Event Type

Action

Class Window

Namespace: OsmiumBedrock

Assembly: Osmium-Bedrock.dll

```
public sealed class Window : GameWindow, IDisposable
```

Inheritance

object ← NativeWindow ← GameWindow ← Window

Implements

IDisposable

Inherited Members

GameWindow.Run() , GameWindow.SwapBuffers() , GameWindow.Close() , GameWindow.TimeSinceLastUpdate() , GameWindow.ResetTimeSinceLastUpdate() , GameWindow.Dispose() , GameWindow.IsMultiThreaded , GameWindow.RenderFrequency , GameWindow.RenderTime , GameWindow.UpdateTime , GameWindow.UpdateFrequency , GameWindow.ExpectedSchedulerPeriod , GameWindow.Load , GameWindow.Unload , GameWindow.UpdateFrame , GameWindow.RenderThreadStarted , GameWindow.RenderFrame , NativeWindow.Focus() , NativeWindow.MakeCurrent() , NativeWindow.ProcessEvents(double) , NativeWindow.ProcessWindowEvents(bool) , NativeWindow.NewInputFrame() , NativeWindow.PointToClient(Vector2i) , NativeWindow.PointToScreen(Vector2i) , NativeWindow.IsKeyDown(Keys) , NativeWindow.IsKeyPressed(Keys) , NativeWindow.IsKeyReleased(Keys) , NativeWindow.IsMouseButtonDown(MouseButton) , NativeWindow.IsMouseButtonPressed(MouseButton) , NativeWindow.IsMouseButtonReleased(MouseButton) , NativeWindow.TryGetCurrentMonitorScale(out float, out float) , NativeWindow.TryGetCurrentMonitorDpi(out float, out float) , NativeWindow.TryGetCurrentMonitorDpiRaw(out float, out float) , NativeWindow.CenterWindow() , NativeWindow.CenterWindow(Vector2i) , NativeWindow.MakeFullscreen(MonitorHandle, int?, int?, int?) , NativeWindow.WindowPtr , NativeWindow.KeyboardState , NativeWindow.JoystickStates , NativeWindow.MousePosition , NativeWindow.MouseState , NativeWindow.IsAnyKeyDown , NativeWindow.IsAnyMouseButtonDown , NativeWindow.VSync , NativeWindow.AutoIconify , NativeWindow.Icon , NativeWindow.IsEventDriven , NativeWindow.ClipboardString , NativeWindow.Title , NativeWindow.API , NativeWindow.Profile , NativeWindow.Flags , NativeWindow.APIVersion , NativeWindow.Context , NativeWindow.CurrentMonitor , NativeWindow.IsFocused , NativeWindow.IsVisible , NativeWindow.Exists ,

NativeWindow.IsExiting , NativeWindow.WindowState , NativeWindow.WindowBorder , NativeWindow.MousePassthrough , NativeWindow.AlwaysOnTop , NativeWindow.Bounds , NativeWindow.Location , NativeWindow.ClientLocation , NativeWindow.Size , NativeWindow.ClientSize , NativeWindow.FramebufferSize , NativeWindow.MinimumSize , NativeWindow.MaximumSize , NativeWindow.AspectRatio , NativeWindow.ClientRectangle , NativeWindow.IsFullscreen , NativeWindow.Cursor , NativeWindow.CursorState , NativeWindow.RawMouseInput , NativeWindow.SupportsRawMouseInput , NativeWindow.HasTransparentFramebuffer , NativeWindow.Move , NativeWindow.Resize , NativeWindow.FramebufferResize , NativeWindow.Refresh , NativeWindow.Closing , NativeWindow.Minimized , NativeWindow.Maximized , NativeWindow.JoystickConnected , NativeWindow.FocusedChanged , NativeWindow.KeyDown , NativeWindow.TextInput , NativeWindow.KeyUp , NativeWindow.MouseLeave , NativeWindow.MouseEnter , NativeWindow.MouseDown , NativeWindow.MouseUp , NativeWindow.MouseMove , NativeWindow.MouseWheel , NativeWindow.FileDrop , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Window()

```
public Window()
```

Namespace OsmiumEditor

Classes

Appdata

Editor

Project

ProjectMemory

ProjectMenu

Class Appdata

Namespace: OsmiumEditor

Assembly: Osmium-Editor.dll

```
public static class Appdata
```

Inheritance

object ← Appdata

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

Path

```
public static readonly string Path
```

Field Value

string

Methods

AssureExistence(string)

```
public static void AssureExistence(string __path)
```

Parameters

__path string

Class Editor

Namespace: OsmiumEditor

Assembly: Osmium-Editor.dll

```
public static class Editor
```

Inheritance

object ← Editor

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

CSProjModuleReferenceEnd

```
public const string CSProjModuleReferenceEnd = "</HintPath>\n
</Reference>\n    </ItemGroup>"
```

Field Value

string

CSProjModuleReferenceMiddle

```
public const string CSProjModuleReferenceMiddle = "\">\n        <HintPath>"
```

Field Value

string

CSProjModuleReferenceStart

```
public const string CSProjModuleReferenceStart = "<ItemGroup>\n
<Reference Include=\""
```

Field Value

string

EditorCompileCSProjEnd

```
public const string EditorCompileCSProjEnd = "\n\n</Project>\n"
```

Field Value

string

EditorCompileCSProjFront

```
public const string EditorCompileCSProjFront = "<Project
Sdk=\"Microsoft.NET.Sdk\">\n\n    <PropertyGroup>\n
<TargetFramework>net10.0</TargetFramework>\n
<ImplicitUsings>enable</ImplicitUsings>\n        <Nullable>enable</Nullable>\n
<OutputPath>.editorCompileOutputPath</OutputPath>\n    </PropertyGroup> \n\n
<ItemGroup>\n        <PackageReference Include=\"OpenTK\" Version=\"4.9.4\" />\n
</ItemGroup> \n   <PropertyGroup>\n\n</PropertyGroup>"
```

Field Value

string

EditorCompileCSProjPath

```
public const string EditorCompileCSProjPath = ".editorCompileCSProj"
```

Field Value

string

EditorCompileOutputPath

```
public const string EditorCompileOutputPath = ".editorCompileOutputPath"
```

Field Value

string

EditorLoadContext

```
public static AssemblyLoadContext EditorLoadContext
```

Field Value

AssemblyLoadContext

EditorModules

```
public static readonly List<Assembly> EditorModules
```

Field Value

List<Assembly>

GlobalUsage

```
public const string GlobalUsage = "//------------------------- READ ME ------------
-------------\n// THIS IS AN AUTO-GENERATED FILE BY OSMIUM! EDITING THIS IS OK,
BUT\n// BE SURE YOU KNOW WHAT YOU ARE DOING, THE AVERAGE USER HAS NO REASON TO BE
HERE\n//-----------------------------------------------------------
\n\n//Automatically references OsmiumNucleus to make life easier!\nglobal using
```

```
OsmiumNucleus;\nglobal using OsmiumBedrock;\n\n//Most IDES will automatically add
this file to avoid redundant usage statements for obvious libraries\n//this is added
to match the user experience of most IDES\nglobal using System;\nglobal using
System.Collections.Generic;\nglobal using System.IO;\nglobal using
System.Linq;\nglobal using System.Net.Http;\nglobal using System.Threading;\nglobal
using System.Threading.Tasks;\n\n//The word IEnumerator can be confusing for
beginners! Coroutine is cleaner and more readable\nglobal using Coroutine =
System.Collections.Generic.IEnumerator<OsmiumNucleus.ICoroutineAction>;"
```

Field Value

string

IDETrickingCSProjEnd

```
public const string IDETrickingCSProjEnd = "\n\n</Project>\n"
```

Field Value

string

IDETrickingCSProjFront

```
public const string IDETrickingCSProjFront = "<Project
Sdk=\"Microsoft.NET.Sdk\">\n\n    <PropertyGroup>\n
<TargetFramework>net10.0</TargetFramework>\n
<ImplicitUsings>enable</ImplicitUsings>\n        <Nullable>enable</Nullable>\n\n
</PropertyGroup> \n\n   <ItemGroup>\n        <PackageReference Include=\"OpenTK\"
Version=\"4.9.4\" />\n    </ItemGroup> \n   \n"
```

Field Value

string

NetVersion

```
public const string NetVersion = "net10.0"
```

Field Value

string

RuntimeCompileCSProjEnd

```
public const string RuntimeCompileCSProjEnd = "\n\n</Project>\n"
```

Field Value

string

RuntimeCompileCSProjFront

```
public const string RuntimeCompileCSProjFront = "<Project
Sdk=\"Microsoft.NET.Sdk\">\n\n    <PropertyGroup>\n
<OutputType>Exe</OutputType>\n        <TargetFramework>net10.0</TargetFramework>\n
<ImplicitUsings>enable</ImplicitUsings>\n        <Nullable>enable</Nullable>\n
<OutputPath>Build/obj</OutputPath>\n    </PropertyGroup> \n\n   <ItemGroup>\n
<PackageReference Include=\"OpenTK\" Version=\"4.9.4\" />\n    </ItemGroup> \n
<PropertyGroup>\n
<BaseIntermediateOutputPath>Build/obj/</BaseIntermediateOutputPath>\n</PropertyGroup
>"
```

Field Value

string

RuntimeCompileCSProjPath

```
public const string RuntimeCompileCSProjPath = ".runtimeCompileCSProj"
```

Field Value

string

RuntimeCompileOutputPath

```
public const string RuntimeCompileOutputPath = "Build/obj"
```

Field Value

string

RuntimeModules

```
public static readonly List<Assembly> RuntimeModules
```

Field Value

List<Assembly>

TopLevelNugget

```
public const string TopLevelNugget = "using
System.Reflection;\n\nOsmium.Initialize();\n\nforeach (AssemblyName assembly in
Assembly.GetExecutingAssembly().GetReferencedAssemblies())\n{\n
Assembly.Load(assembly);\n
Console.WriteLine(assembly.FullName);\n}\n\nBedrock.Run(); \nOsmium.Run();"
```

Field Value

string

TopLevelNuggetPath

```
public const string TopLevelNuggetPath = ".topLevelNugget"
```

Field Value

string

Properties

Assemblies

```
public static List<Assembly> Assemblies { get; }
```

Property Value

List<Assembly>

Modules

```
public static List<Assembly> Modules { get; }
```

Property Value

List<Assembly>

Methods

CompileProject()

```
public static void CompileProject()
```

Main(string[])

```
public static int Main(string[] __args)
```

Parameters

__args string[]

Returns

int

RefreshEditor()

```
public static void RefreshEditor()
```

SaveProject()

```
public static void SaveProject()
```

Events

OnSave

```
public static event Action? OnSave
```

Event Type

Action

Class Project

Namespace: OsmiumEditor

Assembly: Osmium-Editor.dll

```
public static class Project
```

Inheritance

object ← Project

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ProjectPath

```
public static string ProjectPath
```

Field Value

string

Properties

RuntimeModulesPath

```
public static string RuntimeModulesPath { get; }
```

Property Value

string

SourcePath

```
public static string SourcePath { get; }
```

Property Value

string

Methods

GetProjectSubdirectory(bool, params string[])

```
public static string GetProjectSubdirectory(bool regenerate, params
string[] subdirectoryPath)
```

Parameters

`regenerate` bool

`subdirectoryPath` string[]

Returns

string

GetProjectSubpath(bool, params string[])

```
public static string GetProjectSubpath(bool regenerate, params
string[] subdirectoryPath)
```

Parameters

`regenerate` bool

`subdirectoryPath` string[]

Returns

string

Class ProjectMemory

Namespace: OsmiumEditor

Assembly: Osmium-Editor.dll

```
public static class ProjectMemory
```

Inheritance

object ← ProjectMemory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

ProjectRefPath

```
public static string ProjectRefPath
```

Field Value

string

Projects

```
public static List<string> Projects
```

Field Value

List<string>

Methods

AppendProject(string)

```
public static void AppendProject(string __projectPath)
```

Parameters

__projectPath string

ForgetProject(string)

```
public static void ForgetProject(string __projectPath)
```

Parameters

__projectPath string

GetProjectName(string)

```
public static string GetProjectName(string __projectPath)
```

Parameters

__projectPath string

Returns

string

RefreshProjectList()

```
public static void RefreshProjectList()
```

RefreshProjectTime(string)

```
public static void RefreshProjectTime(string __projectPath)
```

Parameters

`__projectPath` string

ValidOsmiumProject(string)

```
public static bool ValidOsmiumProject(string __path)
```

Parameters

`__path` string

Returns

bool

Class ProjectMenu

Namespace: OsmiumEditor

Assembly: Osmium-Editor.dll

```
public static class ProjectMenu
```

Inheritance

object ← ProjectMenu

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

CreateProjectPrompt()

```
public static void CreateProjectPrompt()
```

Initialize()

```
public static void Initialize()
```

OpenProjectPrompt()

```
public static void OpenProjectPrompt()
```

Namespace OsmiumNucleus

Classes

AlwaysUpdate
: Ignores virtualization when receiving events

Debug

DebugMessage

Delay
: Represents a timed delay in a coroutine, which can be used with yield return.

EditorOnly
: Shows that a piece of code is used only for the Editor, and it should be ignored in the compilation phase

Entity
: Any object that transfers updates, and has GameObject children.

GameObject
: The backing class for all Components in Osmium. Inherit this to become one.

OnEditorOpen
: Used to mark any method you would like to call once Osmium opens a project. Only static, parameterless, methods may be called. Your code will compile regardless, but your event will be ignored at runtime.

OnInitialize

Osmium
: Bottom class of Osmium. Carries events from MonoGame into Scenes, and provides OpenTK context.

Scene
: Represents the root in a tree of GameObjects.

Interfaces

ICoroutineAction
: An action taken during a coroutine.

Enums

Event

Represents an Event in Osmium.

MessageType

Class AlwaysUpdate

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

Ignores virtualization when receiving events

```
[AttributeUsage(AttributeTargets.Class)]
public class AlwaysUpdate : Attribute
```

Inheritance

object ← Attribute ← AlwaysUpdate

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) ,
Attribute.GetCustomAttributes(Assembly) , Attribute.GetCustomAttributes(Assembly, bool) ,
Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) ,
Attribute.GetCustomAttributes(MemberInfo) , Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) ,
Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,

object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class Debug

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

```
public static class Debug
```

Inheritance

object ← Debug

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

DefaultCategory

```
public const string DefaultCategory = "GENERAL"
```

Field Value

string

Properties

ThrowErrors

```
public static bool ThrowErrors { get; set; }
```

Property Value

bool

WriteToConsole

```
public static bool WriteToConsole { get; set; }
```

Property Value

bool

Methods

Clear()

```
public static void Clear()
```

CollectVirtualization()

```
public static void CollectVirtualization()
```

Error(object)

```
public static void Error(object Message)
```

Parameters

`Message` object

Error(object, string[], object[])

```
public static void Error(object Message, string[] DetailInfo, object[] Details)
```

Parameters

`Message` object

`DetailInfo` string[]

`Details` object[]

Fatal(object)

```
public static void Fatal(object Message)
```

Parameters

`Message` object

Fatal(object, string[], object[])

```
public static void Fatal(object Message, string[] DetailInfo, object[] Details)
```

Parameters

`Message` object

`DetailInfo` string[]

`Details` object[]

FindCategory(Assembly)

```
public static string FindCategory(Assembly __callingAssembly)
```

Parameters

`__callingAssembly` Assembly

Returns

string

Log(object)

```
public static void Log(object Message)
```

Parameters

`Message` object

Log(object, string[], object[])

```
public static void Log(object Message, string[] DetailInfo, object[] Details)
```

Parameters

`Message` object

`DetailInfo` string[]

`Details` object[]

Message(DebugMessage)

```
public static void Message(DebugMessage message)
```

Parameters

`message` DebugMessage

RegisterCategory(string)

```
public static void RegisterCategory(string __category)
```

Parameters

`__category` string

Warning(object)

```
public static void Warning(object Message)
```

Parameters

`Message` object

Warning(object, string[], object[])

```
public static void Warning(object Message, string[] DetailInfo, object[] Details)
```

Parameters

`Message` object

`DetailInfo` string[]

`Details` object[]

Events

OnClear

```
public static event Action? OnClear
```

Event Type

Action

OnMessage

```
public static event Action<DebugMessage>? OnMessage
```

Event Type

Action<DebugMessage>

Class DebugMessage

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

```
public class DebugMessage
```

Inheritance

object ← DebugMessage

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

DebugMessage(string, string, string[], object[], MessageType)

```
public DebugMessage(string category, string message, string[] detailInfo, object[]
details, MessageType __type)
```

Parameters

`category` string

`message` string

`detailInfo` string[]

`details` object[]

`__type` MessageType

Properties

CallStack

```
public virtual StackTrace CallStack { get; }
```

Property Value

StackTrace

Category

```
public virtual string Category { get; }
```

Property Value

string

DetailInfo

```
public virtual string[] DetailInfo { get; }
```

Property Value

string[]

Details

```
public virtual object[] Details { get; }
```

Property Value

object[]

IsError

```
public virtual bool IsError { get; }
```

Property Value

bool

IsFatal

```
public virtual bool IsFatal { get; }
```

Property Value

bool

IsInfo

```
public virtual bool IsInfo { get; }
```

Property Value

bool

IsWarning

```
public virtual bool IsWarning { get; }
```

Property Value

bool

Message

```
public virtual string Message { get; }
```

Property Value

string

MessageType

```
public virtual MessageType MessageType { get; }
```

Property Value

MessageType

Time

```
public virtual DateTime Time { get; }
```

Property Value

DateTime

Methods

ToFullString()

```
public virtual string ToFullString()
```

Returns

string

ToSimpleString()

```
public virtual string ToSimpleString()
```

Returns

string

ToString()

Returns a string that represents the current object.

```
public override string ToString()
```

Returns

string

A string that represents the current object.

Class Delay

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

Represents a timed delay in a coroutine, which can be used with yield return.

```
public class Delay : ICoroutineAction
```

Inheritance

object ← Delay

Implements

ICoroutineAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Delay(TimeSpan)

Represents a timed delay in a coroutine, which can be used with yield return.

```
public Delay(TimeSpan duration)
```

Parameters

`duration` TimeSpan

The amount of time before the delay ends

Methods

Continue()

Processes the current Action in the coroutine

```
public bool Continue()
```

Returns

bool

Returns false if the coroutine should exit from this action.

FromDays(float)

Creates a new delay from the given time in days

```
public static Delay FromDays(float __days)
```

Parameters

`__days` float

Returns

Delay

FromHours(float)

Creates a new delay from the given time in hours

```
public static Delay FromHours(float __hours)
```

Parameters

`__hours` float

Returns

Delay

FromMilliseconds(float)

Creates a new delay from the given time in milliseconds

```
public static Delay FromMilliseconds(float __milliseconds)
```

Parameters

`__milliseconds` float

Returns

Delay

FromMinutes(float)

Creates a new delay from the given time in minutes

```
public static Delay FromMinutes(float __minutes)
```

Parameters

`__minutes` float

Returns

Delay

FromSeconds(float)

Creates a new delay from the given time in seconds

```
public static Delay FromSeconds(float __seconds)
```

Parameters

`__seconds` float

Returns

Delay

Class EditorOnly

Namespace: OsmiumNucleus

Assembly: Osmium-Editor.dll

Shows that a piece of code is used only for the Editor, and it should be ignored in the compilation phase

```
[AttributeUsage(AttributeTargets.Class|AttributeTargets.Struct|AttributeTargets.Enum
|AttributeTargets.Constructor|AttributeTargets.Method|AttributeTargets.Property|Attr
ibuteTargets.Field|AttributeTargets.Event|AttributeTargets.Interface|AttributeTarget
s.Delegate)]
public class EditorOnly : Attribute
```

Inheritance

object ← Attribute ← EditorOnly

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) ,
Attribute.GetCustomAttributes(Assembly) , Attribute.GetCustomAttributes(Assembly, bool) ,
Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) ,
Attribute.GetCustomAttributes(MemberInfo) , Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) ,
Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,

Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) , Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class Entity

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

Any object that transfers updates, and has GameObject children.

```
public abstract class Entity
```

Inheritance

object ← Entity

Derived

GameObject, Scene

Inherited Members

object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Destroyed

If the Entity has been destroyed.

```
public bool Destroyed { get; }
```

Property Value

bool

Enabled

Whether the Entity is receiving events or not.

```
public bool Enabled { get; }
```

Property Value

bool

Methods

Add(params GameObject[])

Adds every GameObject in Components as children to the Entity.

```
public void Add(params GameObject[] Components)
```

Parameters

`Components` GameObject[]

All the Components to add.

AddScene(Scene)

```
public void AddScene(Scene scene)
```

Parameters

`scene` Scene

AddScene(string)

```
public Scene AddScene(string sceneName)
```

Parameters

`sceneName` string

Returns

Scene

Add<TComponent>()

Adds a new GameObject of type TComponent.

```
public TComponent Add<TComponent>() where TComponent : GameObject, new()
```

Returns

TComponent

The new GameObject that was created.

Type Parameters

`TComponent`

The type of GameObject to add.

Add<TComponent>(string)

Adds a new GameObject of type TComponent with the name Name.

```
public TComponent Add<TComponent>(string Name) where TComponent : GameObject, new()
```

Parameters

`Name` string

The name of the GameObject.

Returns

TComponent

The new GameObject that was created.

Type Parameters

`TComponent`

The type of GameObject to add.

Add<TComponent>(string, params IEnumerable<string>)

Adds a new GameObject of type TComponent and a name of Name with the tags in Tags.

```
public TComponent Add<TComponent>(string Name, params IEnumerable<string> Tags)
where TComponent : GameObject, new()
```

Parameters

`Name` string

The name of the GameObject.

`Tags` IEnumerable<string>

The tags of the new GameObject

Returns

TComponent

The new GameObject that was created.

Type Parameters

`TComponent`

The type of GameObject to create.

Contains(params IEnumerable<GameObject>)

Tests to see if every single GameObject is a direct child of the Entity.

```
public bool Contains(params IEnumerable<GameObject> Components)
```

Parameters

`Components` IEnumerable<GameObject>

The components that are being tested

Returns

bool

True if every GameObject in Components belongs to the Parent, otherwise false.

Contains(params IEnumerable<string>)

Tests to see if the Entity has a child with all the given tags at the very least.

```
public bool Contains(params IEnumerable<string> Tags)
```

Parameters

`Tags` IEnumerable<string>

The list of tags to find on a GameObject

Returns

bool

True if there is at last one GameObject with at least all the given tags, otherwise false.

ContainsScene(Scene)

```
public bool ContainsScene(Scene scene)
```

Parameters

`scene` Scene

Returns

bool

ContainsScene(string)

```
public bool ContainsScene(string sceneName)
```

Parameters

`sceneName` string

Returns

bool

Contains<TComponent>()

Tests if the Entity contains a components of type TComponent.

```
public bool Contains<TComponent>() where TComponent : GameObject, new()
```

Returns

bool

True if the Parent has at least one child of type TComponent, otherwise false.

Type Parameters

`TComponent`

The type of the GameObject you want to check for

Contains<TComponent>(params IEnumerable<string>)

Tests to see if the Entity has a child with at least all the given tags, and of type TComponent.

```
public bool Contains<TComponent>(params IEnumerable<string> Tags) where TComponent :
GameObject, new()
```

Parameters

`Tags` IEnumerable<string>

The list of tags to find on a GameObject

Returns

bool

True if there is at least one GameObject of type TComponent and with all tags in Tags

Type Parameters

`TComponent`

The type of GameObject to search for

DeltaTime()

```
public float DeltaTime()
```

Returns

float

Destroy(params IEnumerable<GameObject>)

Destroys all Components in Components.

```
public void Destroy(params IEnumerable<GameObject> Components)
```

Parameters

`Components` IEnumerable<GameObject>

The list of Components to destroy.

Destroy(params IEnumerable<string>)

Destroys the first found Components that has all the given tags.

```
public void Destroy(params IEnumerable<string> Tags)
```

Parameters

`Tags` IEnumerable<string>

The tags to search for.

DestroyAll()

Destroys all Components owned by the Entity.

```
public void DestroyAll()
```

DestroyAll(params IEnumerable<string>)

Destroys all Components with all the given tags.

```
public void DestroyAll(params IEnumerable<string> Tags)
```

Parameters

`Tags` IEnumerable<string>

The tags to search for.

DestroyAll<TComponent>()

Destroys all Components of type TComponent.

```
public void DestroyAll<TComponent>() where TComponent : GameObject, new()
```

Type Parameters

`TComponent`

The type of GameObject to destroy.

DestroyAll<TComponent>(params IEnumerable<string>)

Destroys all Components of type TComponent with the given tags.

```
public void DestroyAll<TComponent>(params IEnumerable<string> Tags) where TComponent
: GameObject, new()
```

Parameters

`Tags` IEnumerable<string>

The tags to search for.

Type Parameters

`TComponent`

The type of GameObject to destroy.

Destroy<TComponent>()

Destroys the first found Components of type TComponent.

```
public void Destroy<TComponent>() where TComponent : GameObject, new()
```

Type Parameters

`TComponent`

The type of GameObject to destroy.

Destroy<TComponent>(params IEnumerable<string>)

Destroys the first found GameObject of type TComponent with all the given tags.

```
public void Destroy<TComponent>(params IEnumerable<string> Tags) where TComponent :
GameObject, new()
```

Parameters

`Tags` IEnumerable<string>

The tags to search for.

Type Parameters

`TComponent`

The type of component to search for.

Equals(object?)

Determines whether the specified object is equal to the current object.

```
public override bool Equals(object? obj)
```

Parameters

`obj` object

The object to compare with the current object.

Returns

bool

true if the specified object is equal to the current object; otherwise, false.

Get()

Finds the very first GameObject owned by the Entity that belongs to the Entity.

```
public GameObject? Get()
```

Returns

GameObject

The very first GameObject owned by the Entity found in the Entity, null if there is none.

Get(params IEnumerable<string>)

Finds the very first GameObject owned by the Entity that has all the tags in Tags.

```
public GameObject? Get(params IEnumerable<string> Tags)
```

Parameters

`Tags` IEnumerable<string>

The tags to search for

Returns

GameObject

The first GameObject owned by the Entity that has all the tags present in Tags, null if there is none.

GetAll()

Finds all Components that belong to the Entity.

```
public List<GameObject> GetAll()
```

Returns

List<GameObject>

Returns a List<T> of all Components that belongs to the Entity.

GetAll(params IEnumerable<string>)

Finds all Components owned by the Entity that have all the tags in Tags.

```
public List<GameObject> GetAll(params IEnumerable<string> Tags)
```

Parameters

`Tags` IEnumerable<string>

All the tags to search for.

Returns

List<GameObject>

A List<T> of all Components that belong to the Entity and have all the tags in Tags.

GetAll<TComponent>()

Finds all Components owned by the Entity of type TComponent.

```
public List<TComponent> GetAll<TComponent>() where TComponent : GameObject, new()
```

Returns

List<TComponent>

A List<T> of all Components that belong to the Entity and are of type TComponent.

Type Parameters

`TComponent`

The type of GameObject to search for.

GetAll<TComponent>(IEnumerable<string>)

Finds all the Components owned by the Entity of type TComponent that have all the tags in Tags.

```
public List<TComponent> GetAll<TComponent>(IEnumerable<string> Tags) where
TComponent : GameObject, new()
```

Parameters

`Tags` IEnumerable<string>

All the tags to search for.

Returns

List<TComponent>

A List<T> of all Components that belong ot the Entity of type TComponent and have all the tags in Tags.

Type Parameters

`TComponent`

The type of GameObject to search for .

Get<TComponent>()

Finds the very first GameObject owned by the Entity that is of type TComponent.

```
public TComponent? Get<TComponent>() where TComponent : GameObject, new()
```

Returns

TComponent

The very first GameObject owned by the Entity of type TComponent, null if there is none.

Type Parameters

`TComponent`

The true of GameObject to search for

Get<TComponent>(params IEnumerable<string>)

Finds the first GameObject owned by the Entity that is of type TComponent and has all the tags in Tags.

```
public TComponent? Get<TComponent>(params IEnumerable<string> Tags) where TComponent
: GameObject, new()
```

Parameters

`Tags` IEnumerable<string>

The tags to search for

Returns

TComponent

The first GameObject owned by the Entity of type TComponent and has all the tags present in Tags, null if there is none.

Type Parameters

`TComponent`

The type of GameObject to search for

Move(Entity, params IEnumerable<GameObject>)

Moves all the Components in Components to the given Entity.

```
public void Move(Entity entity, params IEnumerable<GameObject> Components)
```

Parameters

`entity` Entity

The docker to move the Components to.

`Components` IEnumerable<GameObject>

the Components to move to the other Entity

Move(Entity, params IEnumerable<string>)

Moves the first found GameObject with all the given tags, to the given Entity.

```
public void Move(Entity entity, params IEnumerable<string> __tags)
```

Parameters

`entity` Entity

The Entity to move the GameObject to.

`__tags` IEnumerable<string>

The tags to search for.

MoveAll(Entity)

Moves all Components to the given Entity.

```
public void MoveAll(Entity entity)
```

Parameters

`entity` Entity

The Entity to move all Components to.

MoveAll(Entity, params IEnumerable<string>)

Moves all Components with the given tags, to the Entity.

```
public void MoveAll(Entity entity, params IEnumerable<string> Tags)
```

Parameters

`entity` Entity

The Entity to move the Components to.

`Tags` IEnumerable<string>

The tags to search for.

MoveAll<TComponent>(Entity)

Moves all Components of type TComponent to the given Entity.

```
public void MoveAll<TComponent>(Entity entity) where TComponent : GameObject, new()
```

Parameters

`entity` Entity

The Entity to move the Components to.

Type Parameters

`TComponent`

The type of Components to move.

MoveAll<TComponent>(Entity, params IEnumerable<string>)

Moves all Components of type TComponent to the Entity.

```
public void MoveAll<TComponent>(Entity entity, params IEnumerable<string> Tags)
where TComponent : GameObject, new()
```

Parameters

`entity` Entity

The Entity to move the Components to.

`Tags` IEnumerable<string>

The tags to search for.

Type Parameters

`TComponent`

The type of GameObject to search for.

Move<TComponent>(Entity)

Moves the first GameObject of type TComponent.

```
public void Move<TComponent>(Entity entity) where TComponent : GameObject, new()
```

Parameters

`entity` Entity

The Entity to move the GameObject to.

Type Parameters

`TComponent`

The type of GameObject to search for.

Move<TComponent>(Entity, params IEnumerable<string>)

Moves the first found GameObject of type TComponent to with all the given tags, to the given Entity.

```
public void Move<TComponent>(Entity entity, params IEnumerable<string> __tags) where
TComponent : GameObject, new()
```

Parameters

`entity` Entity

The Entity to move to the GameObject to.

`__tags` IEnumerable<string>

The tags to search for.

Type Parameters

`TComponent`

The type of GameObject.

RemoveScene(Scene)

```
public void RemoveScene(Scene scene)
```

Parameters

`scene` Scene

RemoveScene(string)

```
public void RemoveScene(string sceneName)
```

Parameters

`sceneName` string

StartCoroutine(IEnumerator<ICoroutineAction>)

```
public void StartCoroutine(IEnumerator<ICoroutineAction> coroutine)
```

Parameters

`coroutine` IEnumerator<ICoroutineAction>

Events

ComponentAdded

```
public static event Action<Entity, GameObject>? ComponentAdded
```

Event Type

Action<Entity, GameObject>

ComponentDestroyed

```
public static event Action<Entity, GameObject>? ComponentDestroyed
```

Event Type

Action<Entity, GameObject>

ComponentMoved

```
public static event Action<Entity, Entity, GameObject>? ComponentMoved
```

Event Type

Action<Entity, Entity, GameObject>

Operators

operator ==(Entity, Entity)

```
public static bool operator ==(Entity __left, Entity __right)
```

Parameters

__left Entity

__right Entity

Returns

bool

operator !=(Entity, Entity)

```
public static bool operator !=(Entity __left, Entity __right)
```

Parameters

__left Entity

__right Entity

Returns

bool

Enum Event

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

Represents an Event in Osmium.

```
public enum Event
```

Fields

`Create = 4`

`Destroy = 5`

`Draw = 3`

`Load = 0`

`Unload = 1`

`Update = 2`

Class GameObject

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

The backing class for all Components in Osmium. Inherit this to become one.

```
public abstract class GameObject : Entity
```

Inheritance

object ← Entity ← GameObject

Inherited Members

Entity.Enabled , Entity.Destroyed , Entity.ComponentMoved , Entity.ComponentAdded , Entity.ComponentDestroyed , Entity.Equals(object) , Entity.Contains(params IEnumerable<GameObject>) , Entity.Contains<TComponent>() , Entity.Contains(params IEnumerable<string>) , Entity.Contains<TComponent>(params IEnumerable<string>) , Entity.Get() , Entity.Get<TComponent>() , Entity.Get(params IEnumerable<string>) , Entity.Get<TComponent>(params IEnumerable<string>) , Entity.GetAll() , Entity.GetAll<TComponent>() , Entity.GetAll(params IEnumerable<string>) , Entity.GetAll<TComponent>(IEnumerable<string>) , Entity.Add(params GameObject[]) , Entity.Add<TComponent>() , Entity.Add<TComponent>(string) , Entity.Add<TComponent>(string, params IEnumerable<string>) , Entity.Destroy(params IEnumerable<GameObject>) , Entity.Destroy<TComponent>() , Entity.Destroy(params IEnumerable<string>) , Entity.Destroy<TComponent>(params IEnumerable<string>) , Entity.DestroyAll() , Entity.DestroyAll<TComponent>() , Entity.DestroyAll(params IEnumerable<string>) , Entity.DestroyAll<TComponent>(params IEnumerable<string>) , Entity.Move(Entity, params IEnumerable<GameObject>) , Entity.Move<TComponent>(Entity) , Entity.Move(Entity, params IEnumerable<string>) , Entity.Move<TComponent>(Entity, params IEnumerable<string>) , Entity.MoveAll(Entity) , Entity.MoveAll<TComponent>(Entity) , Entity.MoveAll(Entity, params IEnumerable<string>) , Entity.MoveAll<TComponent>(Entity, params IEnumerable<string>) , Entity.AddScene(Scene) , Entity.AddScene(string) , Entity.RemoveScene(Scene) , Entity.RemoveScene(string) , Entity.ContainsScene(Scene) , Entity.ContainsScene(string) , Entity.StartCoroutine(IEnumerator<ICoroutineAction>) , Entity.DeltaTime() , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

Tags

A list of all tags belonging to the component. Use AddTag() to modify it. Easy to calculate, but setting this requires that the Parent resorts Components;

```
public HashSet<string> Tags
```

Field Value

HashSet<string>

Properties

Name

The GameObject's name

```
public string Name { get; set; }
```

Property Value

string

Parent

Current parent of the GameObject. Can be either a Scene or another GameObject.

```
public Entity Parent { get; }
```

Property Value

Entity

Scene

Scene the GameObject resides in. To save ram it's not stored but calculated on the fly.

```
public Scene Scene { get; }
```

Property Value

Scene

Methods

Destroy()

Makes the GameObject destroy itself

```
public void Destroy()
```

Move(Entity)

```
public void Move(Entity __newDocker)
```

Parameters

`__newDocker` Entity

ToString()

Just gives the GameObject's name. But makes debugging a little easier :).

```
public override string ToString()
```

Returns

string

Interface ICoroutineAction

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

An action taken during a coroutine.

```
public interface ICoroutineAction
```

Methods

Continue()

Processes the current Action in the coroutine

```
bool Continue()
```

Returns

bool

Returns false if the coroutine should exit from this action.

Enum MessageType

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

```
[Flags]
public enum MessageType : byte
```

Fields

`Error = 4`

`Fatal = 8`

`Info = 1`

`Warning = 2`

Class OnEditorOpen

Namespace: OsmiumNucleus

Assembly: Osmium-Editor.dll

Used to mark any method you would like to call once Osmium opens a project. Only static, parameterless, methods may be called. Your code will compile regardless, but your event will be ignored at runtime.

```
[AttributeUsage(AttributeTargets.Method)]
public class OnEditorOpen : Attribute
```

Inheritance

object ← Attribute ← OnEditorOpen

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) ,
Attribute.GetCustomAttributes(Assembly) , Attribute.GetCustomAttributes(Assembly, bool) ,
Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) ,
Attribute.GetCustomAttributes(MemberInfo) , Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) ,
Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,

Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Class OnInitialize

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

```
[AttributeUsage(AttributeTargets.Method)]
public class OnInitialize : Attribute
```

Inheritance

object ← Attribute ← OnInitialize

Inherited Members

Attribute.Equals(object) , Attribute.GetCustomAttribute(Assembly, Type) ,
Attribute.GetCustomAttribute(Assembly, Type, bool) ,
Attribute.GetCustomAttribute(MemberInfo, Type) ,
Attribute.GetCustomAttribute(MemberInfo, Type, bool) ,
Attribute.GetCustomAttribute(Module, Type) , Attribute.GetCustomAttribute(Module, Type, bool) ,
Attribute.GetCustomAttribute(ParameterInfo, Type) ,
Attribute.GetCustomAttribute(ParameterInfo, Type, bool) ,
Attribute.GetCustomAttributes(Assembly) , Attribute.GetCustomAttributes(Assembly, bool) ,
Attribute.GetCustomAttributes(Assembly, Type) ,
Attribute.GetCustomAttributes(Assembly, Type, bool) ,
Attribute.GetCustomAttributes(MemberInfo) , Attribute.GetCustomAttributes(MemberInfo, bool) ,
Attribute.GetCustomAttributes(MemberInfo, Type) ,
Attribute.GetCustomAttributes(MemberInfo, Type, bool) , Attribute.GetCustomAttributes(Module) ,
Attribute.GetCustomAttributes(Module, bool) , Attribute.GetCustomAttributes(Module, Type) ,
Attribute.GetCustomAttributes(Module, Type, bool) ,
Attribute.GetCustomAttributes(ParameterInfo) ,
Attribute.GetCustomAttributes(ParameterInfo, bool) ,
Attribute.GetCustomAttributes(ParameterInfo, Type) ,
Attribute.GetCustomAttributes(ParameterInfo, Type, bool) , Attribute.GetHashCode() ,
Attribute.IsDefaultAttribute() , Attribute.IsDefined(Assembly, Type) ,
Attribute.IsDefined(Assembly, Type, bool) , Attribute.IsDefined(MemberInfo, Type) ,
Attribute.IsDefined(MemberInfo, Type, bool) , Attribute.IsDefined(Module, Type) ,
Attribute.IsDefined(Module, Type, bool) , Attribute.IsDefined(ParameterInfo, Type) ,
Attribute.IsDefined(ParameterInfo, Type, bool) , Attribute.Match(object) , Attribute.TypeId ,
object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object) , object.ToString()

Constructors

OnInitialize(bool)

```
public OnInitialize(bool __ignoreVirtualization = false)
```

Parameters

`__ignoreVirtualization` bool

Fields

IgnoreVirtualization

```
public readonly bool IgnoreVirtualization
```

Field Value

bool

Class Osmium

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

Bottom class of Osmium. Carries events from MonoGame into Scenes, and provides OpenTK context.

```
public static class Osmium
```

Inheritance

object ← Osmium

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

SafeEscape

If Osmium crashes, with SafeEscape enabled, it will still allow exiting statements to be run

```
public static bool SafeEscape
```

Field Value

bool

Version

Current Version of Osmium

```
public const string Version = "1.3"
```

Field Value

string

Properties

Assemblies

All assemblies loaded by Osmium

```
public static IReadOnlyList<Assembly> Assemblies { get; }
```

Property Value

IReadOnlyList<Assembly>

DeltaTime

The seconds since the last type of frame event. For instance if Draw was just called, DeltaTime would currently reflect the time since the last DRAW call, even though there was an Update call in between. Use this for framerate independent logic!

```
public static float DeltaTime { get; }
```

Property Value

float

IsClosed

Displays if Osmium has closed or not.

```
public static bool IsClosed { get; }
```

Property Value

bool

IsInitialized

Displays if Osmium has Started or not.

```
public static bool IsInitialized { get; }
```

Property Value

bool

IsRunning

Displays if the update loop is active.

```
public static bool IsRunning { get; }
```

Property Value

bool

IsVirtualized

Displays if Osmium is running virtually

```
public static bool IsVirtualized { get; }
```

Property Value

bool

Scenes

List of all scenes currently loaded in the kernel.

```
public static IReadOnlySet<Scene> Scenes { get; }
```

Property Value

IReadOnlySet<Scene>

Window

Bottom class of Osmium. Contains the OpenTK Instance.

```
public static Window Window { get; }
```

Property Value

Window

Methods

AddScene(Scene)

Adds a new Scene to Osmium!

```
public static void AddScene(Scene __scene)
```

Parameters

`__scene` Scene

The Scene you want to add

Remarks

Using this is only recommended for custom types of Scene! If you wish to Spawn in a normal scene, use AddScene(string)

AddScene(string)

Creates a new Scene with the given name, returns null if creating it fails

```
public static Scene? AddScene(string __name)
```

Parameters

`__name` string

The name of the new Scene and its unique identifier as a string; cannot match up with any other names in the scene.

Returns

Scene

The Scene you created, or null if there is a failure.

Close()

Closes Osmium and ends the OpenTK instance!

```
public static void Close()
```

ContainsScene(Scene)

Tells you whether a Scene is currently loaded or not

```
public static bool ContainsScene(Scene __scene)
```

Parameters

`__scene` Scene

A reference to the Scene you are looking for

Returns

bool

A bool that is either true or false depending on if a scene with the given name is loaded or not

ContainsScene(string)

Tells you whether a Scene is currently loaded or not

```
public static bool ContainsScene(string __name)
```

Parameters

__name string

The name of the Scene you are searching for

Returns

bool

A bool that is either true or false depending on if a scene with the given name is loaded or not

EditorInitialize()

Initializes the Context and marks Osmium as initialized; but does not Resolve types

```
public static void EditorInitialize()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. These methods are made required in order to use Virtualization! Use Editor Methods instead of normal ones for Virtualization to work.

EditorRun()

Starts OpenTK but doesn't let the update loop run!

```
public static void EditorRun()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. These methods are made required in order to use Virtualization! Use Editor Methods instead of normal ones for Virtualization to work.

GetScene(string)

Finds a Scene from the given name.

```
public static Scene? GetScene(string __name)
```

Parameters

__name string

The name of the Scene you want to find

Returns

Scene

The newly found Scene! Or null if there currently isn't one

Initialize()

Starts up Osmium, and resolves all the Components in the App Domain! Must be called before Run() and Osmium is not guaranteed to work until you call this method!

```
public static void Initialize()
```

RemoveScene(Scene)

Removes a currently loaded Scene from Osmium

```
public static void RemoveScene(Scene __scene)
```

Parameters

__scene Scene

The scene to remove

RemoveScene(string)

Removes a currently loaded Scene from Osmium from a given name

```
public static void RemoveScene(string __name)
```

Parameters

__name string

The name of the scene you want to remove

Run()

Begins the update loop and sends Load() to all Components!

```
public static void Run()
```

StartCoroutine(IEnumerator<ICoroutineAction>)

Starts a coroutine

```
public static void StartCoroutine(IEnumerator<ICoroutineAction> __coroutine)
```

Parameters

__coroutine IEnumerator<ICoroutineAction>

VirtualClose()

Pretends to close Osmium, and makes the Components think that the Game has ended.

```
public static void VirtualClose()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for the Editor! So don't use it unless you know what you are doing. If you do want to use it, use the EditorInitialize() EditorRun() and EditorClose() instead of the traditional methods!

VirtualInitialize(IEnumerable<Assembly>)

Pretends to initialize Osmium, and makes the Components think that the Game has just been initialized.

```
public static void VirtualInitialize(IEnumerable<Assembly> __assemblies)
```

Parameters

`__assemblies` IEnumerable<Assembly>

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. If you do want to use it, use the EditorInitialize() EditorRun() and EditorClose() instead of the traditional methods!

VirtualRun()

Pretends to run Osmium virtually, and makes the Components think it has Started.

```
public static void VirtualRun()
```

Remarks

This is part of the Editor pipeline! It has no error checking, and it is made explicitly for Radium! So don't use it unless you know what you are doing. If you do want to use it, use the EditorInitialize() EditorRun() and EditorClose() instead of the traditional methods!

Events

Draw

```
public static event Action? Draw
```

Event Type

Action

FinalDraw

An event that is raised after all draw calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalDraw
```

Event Type

Action

FinalLoad

An event that is raised after all load calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalLoad
```

Event Type

Action

FinalUnload

An event that is raised after all unload calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalUnload
```

Event Type

Action

FinalUpdate

An event that is raised after all update calls, this is meant to be used for libraries that require overhead

```
public static event Action? FinalUpdate
```

Event Type

Action

FirstDraw

An event that is raised before all draw calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstDraw
```

Event Type

Action

FirstLoad

An event that is raised before all load calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstLoad
```

Event Type

Action

FirstUnload

An event that is raised before all unload calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstUnload
```

Event Type

Action

FirstUpdate

An event that is raised before all update calls, this is meant to be used for libraries that require overhead

```
public static event Action? FirstUpdate
```

Event Type

Action

Load

```
public static event Action? Load
```

Event Type

Action

SceneAdded

Is called when a new Scene is added!

```
public static event Action<Scene>? SceneAdded
```

Event Type

Action<Scene>

SceneRemoved

Is called when a new Scene is removed!

```
public static event Action<Scene>? SceneRemoved
```

Event Type

Action<Scene>

Unload

```
public static event Action? Unload
```

Event Type

Action

Update

```
public static event Action? Update
```

Event Type

Action

Class Scene

Namespace: OsmiumNucleus

Assembly: Osmium-Nucleus.dll

Represents the root in a tree of GameObjects.

```
public class Scene : Entity
```

Inheritance

object ← Entity ← Scene

Inherited Members

Entity.Enabled , Entity.Destroyed , Entity.ComponentMoved , Entity.ComponentAdded , Entity.ComponentDestroyed , Entity.Equals(object) , Entity.Contains(params IEnumerable<GameObject>) , Entity.Contains<TComponent>() , Entity.Contains(params IEnumerable<string>) , Entity.Contains<TComponent>(params IEnumerable<string>) , Entity.Get() , Entity.Get<TComponent>() , Entity.Get(params IEnumerable<string>) , Entity.Get<TComponent>(params IEnumerable<string>) , Entity.GetAll() , Entity.GetAll<TComponent>() , Entity.GetAll(params IEnumerable<string>) , Entity.GetAll<TComponent>(IEnumerable<string>) , Entity.Add(params GameObject[]) , Entity.Add<TComponent>() , Entity.Add<TComponent>(string) , Entity.Add<TComponent>(string, params IEnumerable<string>) , Entity.Destroy(params IEnumerable<GameObject>) , Entity.Destroy<TComponent>() , Entity.Destroy(params IEnumerable<string>) , Entity.Destroy<TComponent>(params IEnumerable<string>) , Entity.DestroyAll() , Entity.DestroyAll<TComponent>() , Entity.DestroyAll(params IEnumerable<string>) , Entity.DestroyAll<TComponent>(params IEnumerable<string>) , Entity.Move(Entity, params IEnumerable<GameObject>) , Entity.Move<TComponent>(Entity) , Entity.Move(Entity, params IEnumerable<string>) , Entity.Move<TComponent>(Entity, params IEnumerable<string>) , Entity.MoveAll(Entity) , Entity.MoveAll<TComponent>(Entity) , Entity.MoveAll(Entity, params IEnumerable<string>) , Entity.MoveAll<TComponent>(Entity, params IEnumerable<string>) , Entity.AddScene(Scene) , Entity.AddScene(string) , Entity.RemoveScene(Scene) , Entity.RemoveScene(string) , Entity.ContainsScene(Scene) , Entity.ContainsScene(string) , Entity.StartCoroutine(IEnumerator<ICoroutineAction>) , Entity.DeltaTime() , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Scene(string)

Represents the root in a tree of GameObjects.

```
public Scene(string Name)
```

Parameters

`Name` string

The unique identifier of the Scene

Properties

Name

The fully unique identifier of the Scene. Must not match a name from any other Scene. And cannot be null.

```
public string Name { get; set; }
```

Property Value

string

Namespace Osmium_Tests

Classes

Tests

Class Tests

Namespace: Osmium_Tests

Assembly: Osmium-Tests.dll

```
[SingleThreaded]
[NonParallelizable]
public class Tests
```

Inheritance

object ← Tests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Setup()

```
[SetUp]
public void Setup()
```

Test1()

```
[Test]
public void Test1()
```